

Project ID: BTP2CSE210

Brain Tumor Detection

MOBILE-INTEGRATED NEURAL DIAGNOSTIC SYSTEM
(TUMORAI)

TECHNICAL PROJECT SYNOPSIS

AI-Powered Oncology & Healthcare Informatics

Prepared By: Engineering Team

Domain: Computer Vision & Medical Diagnostics

Date: March 2026

Table of Contents

1. Introduction	1
2. Problem Statement	2
3. Literature Review	3
4. Objectives	4
5. Methodology	5
6. Tools & Technologies	7
7. Expected Outcomes	8
8. Applications	9
9. Conclusion and Future Work	10

1. Introduction

The integration of Artificial Intelligence (AI) into clinical workflows has revolutionized the speed and accuracy of neurological assessments. Brain tumors, characterized by the abnormal growth of cells within the cranium, require precision diagnostics to optimize surgical outcomes and patient recovery. While MRI remains the gold standard for neuroimaging, the sheer volume of data often results in diagnostic backlogs.

Brain Tumor Detection (TumorAi) is developed as a high-performance, mobile-first screening tool. By utilizing On-Device Machine Learning (ODML), the platform empowers healthcare providers to analyze MRI scans in real-time without the need for high-latency cloud processing or expensive desktop hardware. This project focuses on democratizing specialized medical diagnostics through accessible mobile technology.

2. Problem Statement

Neurological diagnostics currently face several structural challenges that impact patient care:

- **Expert Availability:** Senior neuroradiologists are often concentrated in urban medical centers, leaving rural clinics with significant gaps in diagnostic expertise.
- **Diagnostic Latency:** Traditional reporting cycles can take hours or days, which is suboptimal for aggressive or emergency tumor cases.
- **Cognitive Load:** Manual review of hundreds of slices per scan is prone to human fatigue, leading to potential missed micro-calcifications or subtle masses.

The project addresses these issues by providing an automated, standardized "second opinion" that can flag suspicious scans instantly for prioritized specialist review.

3. Literature Review

Recent advancements in Convolutional Neural Networks (CNNs) have established a new benchmark for automated tumor detection. Architectures such as MobileNetV2 and EfficientNet have proven that high-accuracy feature extraction can be achieved even on computationally constrained devices.

Prior research indicates that models trained on curated datasets like the *BraTS (Brain Tumor Segmentation)* challenge achieve competitive sensitivity and specificity. Harboring a weight-quantized TFLite model, this project transitions these research-grade algorithms into a field-ready mobile application, optimizing for both inference speed and battery efficiency.

4. Objectives

The core objectives of the implementation include:

- **Mobile-First Inference:** Deploying a pre-trained CNN model using TensorFlow Lite for seamless local execution.
- **High sensitivity:** Achieving reliable classification outputs that assist in identifying potential malignancies.
- **Persistent Logging:** Utilizing Firebase Firestore to maintain a digital history of scans for longitudinal patient analysis.
- **Clinical UX:** Designing a minimal, distraction-free interface that provides clear confidence metrics and actionable results.

5. Methodology

The implementation follows a rigorous technical pipeline to ensure diagnostic consistency:

1. Dynamic Preprocessing:

Input MRI images are captured via the Android UI and processed through an optimized pipeline. This includes resizing to 224x224 and pixel normalization to ensure the model receives standardized input tensors.

2. TFLite Interpreter Integration:

A dedicated TFLite helper manages memory-mapped model files and execution threads. By using delegated CPU/GPU inference, the application ensures the main UI thread remains responsive during heavy computation.

3. Probabilistic Classification:

The model yields a raw floating-point score, which is interpreted through a softmax or thresholding layer to produce the final classification: "Tumor Detected" or "No Tumor".

4. Cloud Data Sync:

Scan results are automatically serialized and pushed to a secure cloud-based Firestore collection, timestamped for historical reference.

6. Tools & Technologies

LANGUAGE

Kotlin (1.9+)

PLATFORM

Android SDK (API 34)

ML ENGINE

TensorFlow Lite

BACKEND

Firebase Firestore

BINDING

ViewBinding

CONCURRENCY

Kotlin Coroutines

7. Expected Outcomes

The project aims to deliver a tool that offers:

- Instantaneous (sub-500ms) classification of MRI images.
- Reduction in the barrier to entry for preliminary neuro-screenings.
- A scalable architecture that can be extended to other types of medical imaging (X-Ray, CT).

8. Applications

The project finds utility across the healthcare spectrum:

- **Emergency Rooms:** Rapidly flagging potential brain trauma or tumors during initial intake.
- **Diagnostic Laboratories:** Assisting technicians in quality control and preliminary report generation.
- **Tele-Medicine:** Enabling remote health workers to provide initial insights before patient transport.

9. Conclusion and Future Work

Brain Tumor Detection exemplifies the synergy between mobile software engineering and artificial intelligence. By bringing advanced diagnostic capabilities to the edge, we enhance the tools available to medical professionals worldwide.

Future Roadmap: We plan to integrate *Grad-CAM* visualizations to provide clinicians with a "heatmap" of the model's focus. Additionally, we aim to extend detection to specific tumor grades (Low Grade vs. High Grade Gliomas) to further refine clinical decision-making.